

Lesson 2: Blast Off Into Scratch — Setup and Your First Sprite

Big Idea

Scratch is our coding playground, and a sprite is a character we bring to life by snapping together colorful blocks like puzzle pieces.

Learning Objectives

- Open the Scratch editor and identify its main areas (stage, block palette, code area, sprite list).
- Understand what a "sprite" and a "block" are.
- Drag and snap at least one block into the code area and run it by clicking it.
- Add a new sprite from the sprite library and give it a costume.
- Save their project (or understand how the teacher will save it for them).

Teacher Preparation

- Ensure every device can reach the Scratch editor (<https://scratch.mit.edu/> — "Create" button, no login required to build, though an account helps to save online).
- Test that clicking a block plays a sound/does an action, and that the green flag works, on your classroom devices beforehand.
- Have a simple "my first project" example ready to show onscreen.
- Remind students of Lesson 1: "Remember how WE had to give super clear instructions to our robot? Today, Scratch blocks ARE those instructions — but this time, snapped together like puzzle pieces!"

Required Files

- [Student_Lesson_Guide.md](#)
- [Activities.md](#)
- [Challenges.md](#)

Teaching Flow (approx. 45 minutes)

1. Warm-up / Hook — 4 min
2. Concept Introduction — 8 min
3. Teacher Demonstration — 8 min
4. Guided Coding — 10 min
5. Mini Challenge — 5 min
6. Main Activity — 8 min
7. Check for Understanding + Recap — 2 min

Warm-up / Hook (4 min)

 **Coding Mission:** "Remember last time, we became robots and coders without a computer? Today we get to open a real coding tool called Scratch, where our instructions become blocks we can literally snap together, like Lego bricks — and today, YOU get to bring a character to life for the first time!"

Concept Introduction (8 min)

- **The Stage** — like a little TV screen where our story or game happens.
- **Sprites** — the characters or objects in our project (a cat, a dog, a rocket ship — anything!). Explain: "A sprite is like an actor. It can move, talk, and change costumes, but only if we give it code."
- **Blocks** — colorful puzzle pieces, each one a single instruction (like "move," "say," or "play sound"). Different colors mean different types of instructions (motion blocks are blue, looks blocks are purple, sound blocks are pink/magenta).
- **The Code Area** — the empty space where we snap blocks together to build a script (a stack of instructions).
-  **Robot Tip:** "Remember our robot game? Each Scratch block is one clear instruction — just like 'take 3 steps forward.' When you snap blocks together, you're writing a recipe the sprite will follow exactly."

Teacher Demonstration (8 min)

1. Open Scratch, point out: Stage (top left/center), Sprite list (bottom), Block Palette (middle), Code Area (right).
2. Click on the built-in Cat sprite. Say: "This is our very first sprite!"
3. Drag a "motion_movesteps" block into the code area.
4. Click directly on the block — show that the cat moves! Say: "See that? Clicking a block runs it immediately, just like a test button."

5. Snap a "looks_say" block underneath, type "Hello, I'm a sprite!" Click the stack — show the cat says it.
6. Show how to add a NEW sprite: click the sprite button (bottom right, cat-face icon with a plus), choose one from the library.
7. Show costumes tab briefly: "Look — this sprite even has different outfits called costumes!"

Guided Coding (10 min)

Students, on their own devices (or in pairs), open Scratch and: 1. Find and click on the default Cat sprite. 2. Drag a "move 10 steps" block into the code area. 3. Click it a few times and watch the cat move across the stage. 4. Add a "say Hello for 2 seconds" block underneath it, and click the stack to run it.

Mini Challenge (5 min)

Add ONE more block of your choice (any color!) underneath your stack, and figure out — just by clicking it and watching — what it does. Be ready to explain it to a partner in your own words.

Main Activity / Project (8 min)

Add and Meet a New Sprite: Students click the "add sprite" button and choose a brand-new sprite from the library (an animal, a food, a fantasy character — anything they like). They: 1. Rename their sprite (click the name field in the sprite info area, type a fun new name). 2. Give it a "move" block and a "say" block with a message that introduces the sprite (e.g., "Hi, I'm Sparkle the Unicorn!"). 3. Click the green flag to test their whole stack together.

Fun Challenge (optional, harder) ★

Add a second costume change to your sprite's script using a "next costume" block, and try adding a sound block too, so your sprite moves, talks, changes look, AND makes a sound — all from clicking one stack!

Check for Understanding

- "What is a sprite?"
- "What is a block, and what happens when you click on it by itself?"
- "Where on the screen do you see the sprite actually move or talk?"
- "How do you add a brand-new sprite to your project?"

Common Student Mistakes (with fixes)

- **Mistake:** Blocks don't snap together (there's a gap). **Fix:** Show them to drag the new block so its top notch is right under the bump of the block above — it will highlight with a white outline when it's close enough to snap.
- **Mistake:** Clicking the block palette instead of dragging. **Fix:** Remind them: click AND HOLD, then drag into the code area, then let go.
- **Mistake:** Forgetting which sprite they're coding for (blocks appear to "not work"). **Fix:** Check the sprite list — whichever sprite is highlighted/selected is the one being coded right now.
-  **Debug Detective:** "Uh-oh! A student's cat sprite isn't moving when clicked, even though there's a 'move 10 steps' block in the code area. What's wrong?" (Answer: the block isn't actually clicked/selected sprite might be different, or the block isn't connected to anything and needs to be clicked directly, or the sprite is off-stage. Walk through checking: Is the right sprite selected? Is the block by itself clickable? Try clicking it directly.)

Differentiation

- **Struggling support:** Pair them with a partner; focus only on move + say blocks; teacher pre-selects the new sprite from the library for them.
- **Standard:** Follow full lesson as planned.
- **Extension for advanced learners:** Have them add TWO sprites, each with its own move+say script, and predict out loud what will happen when they click the green flag for the whole project (both sprites act at once!).

Recap

"Today you opened Scratch for the very first time, met your first sprite, and code by snapping blocks together like puzzle pieces. You made a sprite move, talk, and even added a brand-new character of your own!"

Take-Home / Creative Challenge

Think of a sprite you'd love to bring to life next time (a dragon, a soccer ball, your pet) — sketch what it might look like and one silly thing you'd want it to say.

Teacher Talking Points

- "Look at that — you just wrote your very first line of real code, and it worked!"
- "It's totally okay if a block doesn't do what you expected. That's not 'breaking' anything — that's exploring!"
- "Every single block is like one instruction from our robot game — you already know how this works."

- "I can already tell some amazing sprites are about to come to life in this room."



Victory Moment

Have every student click their green flag at the same time so the whole room's sprites move and talk together.

Celebrate: "You just ran real code, together, as a whole class of coders!"